

School of Artificial Intelligence and Automation
Huazhong University of Science and Technology
Wuhan 430074, P.R. China
wangliang.f@gmail.com

May 30, 2026

The Editors
Ergodic Theory and Dynamical Systems
Cambridge University Press

Dear Editors,

I am submitting the enclosed manuscript, *Transient Chaos and Topological Bounds in Prime Dynamics: Revisiting the One-Dimensional Sieve Mapping*, for consideration as a research article.

The paper studies the symbolic dynamics of the Eratosthenes sieve under the Metropolis–Stein–Stein (MSS) kneading admissibility framework, taking as starting object the recently proposed matching between the sieve word Q_k and admissible itineraries of the unimodal map $x \mapsto 1 - ux^2$ at the band-merging parameter $u_c \approx 1.5436890$. Two unconditional theorems anchor the contribution. The *Parity-Gap Lemma* (Theorem A) shows that any MSS-defective shift of $W_k = Q_k[0, p_{k+1}^2)$ forces a prime gap of length $\geq p_{k+1} - 1$ at a specific position; this reduces the topological-admissibility question to an extremal prime-gap inequality, and we observe that classical gap bounds (Legendre, Andrica, RH, BHP) all fail to clear it—only a strongly sub-root bound such as Cramér’s conjecture forces eventual admissibility. The *even-gap rigidity* theorem (Theorem B) shows that gaps between consecutive L -symbols at u_c are almost surely even, via the parity-twisted lex order on RLR^∞ . A third structural result (Theorem C) shows the gap measure decays asymptotically geometrically with ratio $p_\infty \approx 0.596$, ruling out internal mod-3 resonance and identifying the unimodal projection as an abelian / mod-2 shadow of the prime distribution. The paper closes with a clearly conditional reduction connecting recurrence to the L - R - L cylinder to a precisely stated arithmetic shadowing problem; we explicitly disclaim any consequence for classical number-theoretic conjectures.

The paper fits squarely within the journal’s scope, intersecting one-dimensional kneading theory, ergodic averages on non-uniformly hyperbolic systems, and arithmetic dynamics in the spirit of Cellarosi–Sinai (*ETDS*, 2013). The proofs rely on classical results of Misiurewicz, Mañé, Lasota–Yorke, and Young.

I should disclose that I am also the author of a separate manuscript (currently under review elsewhere) that proposes the unimodal–sieve matching itself; the present paper independently analyses the admissibility structure of that matching and is structured to stand alone—no theorem statement depends on the prior work, as recorded in the title-page footnote. The manuscript is not under consideration elsewhere; there are no competing interests; and the complete LaTeX source, the numerical experiments, the figure-generation scripts, and six companion Jupyter notebooks reproducing the headline numerical claims are openly available at https://github.com/maris205/prime_dynamics.

I thank the Editors and referees in advance for their consideration.

Sincerely yours,

Liang Wang